Research Paper Explorer — Rapport

Question : How can retrieval reduce hallucinations in LLMs?

Généré le 24/10/2026 à 12:00

3 papier(s) analysé(s)

Réponse

Retrieval reduces hallucinations by grounding the generator in retrieved passages [arXiv:2401.00001]. Consistency checks between several sampled answers can also flag unsupported statements without external knowledge [arXiv:2403.00002].

Calibrating the retriever's scores (temperature α) further improves exact match.

Description des papiers

Retrieval-Augmented Generation for Open-Domain QA with α-Calibrated Retrievers

Zoë Müller, José Núñez, Łukasz Kowalski et al. · 2024 · ACL · arXiv:2401.00001 · pertinence : élevée

Problème : Reducing hallucinations of LLMs on open-domain question answering.

Méthode : A dense retriever feeds top-k passages to a generator; retrieval scores are calibrated with a temperature α learned on a validation set.

Datasets & métriques : Natural Questions, TriviaQA — métriques : EM, F1

Résultats : EM 45.2 on Natural Questions (+3.1 over the baseline).

Limitations : English only; Retriever trained on Wikipedia

Self-Checking Generation: Detecting Hallucinations without External Knowledge

Hélène Dubois, Kenji Sato · 2024 · EMNLP Findings · arXiv:2403.00002 · pertinence : moyenne

Problème : Detecting hallucinated statements in generated text.

Méthode : Sample several answers and measure their consistency (self-consistency).

Datasets & métriques : WikiBio — métriques : AUC-PR

Résultats : Non mentionné dans le résumé

Limitations : Non mentionné dans le résumé

Speech Translation for Low-Resource Languages

Amina Diallo · 2025 · arXiv:2505.00003 · pertinence : peu pertinent

Problème : Speech translation for languages with little data.

Méthode : Cascade ASR + MT with data augmentation.

Datasets & métriques : CoVoST 2 — métriques : BLEU

Résultats : Non mentionné dans le résumé

Limitations : Only 3 language pairs

Tableau comparatif

Papier	Méthode	Datasets	Résultats	Limitations	Pertinence
arXiv:2401.00001 Retrieval-Augmented Generation for Open-Domain QA with α-Calibrated Retrievers (2024)	A dense retriever feeds top-k passages to a generator; retrieval scores are calibrated with a temperature α learned on a validation set.	Natural Questions, TriviaQA	EM 45.2 on Natural Questions (+3.1 over the baseline).	English only; Retriever trained on Wikipedia	élevée
arXiv:2403.00002 Self-Checking Generation: Detecting Hallucinations without External Knowledge (2024)	Sample several answers and measure their consistency (self-consistency).	WikiBio	Non mentionné dans le résumé	Non mentionné dans le résumé	moyenne
arXiv:2505.00003 Speech Translation for Low-Resource Languages (2025)	Cascade ASR + MT with data augmentation.	CoVoST 2	Non mentionné dans le résumé	Only 3 language pairs	peu pertinent

Synthèse comparative

Both papers address hallucinations, one by grounding generation in retrieved passages [arXiv:2401.00001] and the other by checking self-consistency [arXiv:2403.00002]. They do not share datasets. The retrieval-based approach seems best suited to the question because it reports a gain on an open-domain QA benchmark [arXiv:2401.00001]. The third paper is about speech translation and is only marginally related [arXiv:2505.00003]. Limitations are rarely stated in the abstracts.

Références

1 [arXiv:2401.00001] Retrieval-Augmented Generation for Open-Domain QA with α-Calibrated Retrievers. Zoë Müller, José Núñez, Łukasz Kowalski et al., 2024. https://arxiv.org/abs/2401.00001

2 [arXiv:2403.00002] Self-Checking Generation: Detecting Hallucinations without External Knowledge. Hélène Dubois, Kenji Sato, 2024. https://arxiv.org/abs/2403.00002

3 [arXiv:2505.00003] Speech Translation for Low-Resource Languages. Amina Diallo, 2025. https://arxiv.org/abs/2505.00003